

## WEEK 07 · REFERENCE

# THE RULES, IN ONE PLACE

Thresholds, decision trees and the three scales. Keep this one; it answers the questions you will actually have in week 11.

## WHAT IS IN HERE

Every threshold this week names, the four states with their failure modes, and the three scales.

## HOW TO READ IT

First column is the case you are in. The last one is what it costs, or what to do.

## WHEN TO OPEN IT

Mid-build, when a state, a focus stop or a contrast ratio has become a guess.

► **THE WORKING TOOL IS ELSEWHERE:** `audit-checklist-he.pdf` — fifty-five rows, printable. This deck is the reasoning behind them.

# CONTRAST: THREE NUMBERS, AND WHAT EACH COVERS

| RATIO   | APPLIES TO                                                                | WHERE PEOPLE GET IT WRONG                                                       |
|---------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 4.5 : 1 | Normal text — under 24px, or under 18.66px if bold                        | Placeholder text and disabled labels are still text                             |
| 3 : 1   | Large text — 24px, or 18.66px bold (18pt, 14pt bold)                      | Assuming a big heading is exempt. It is not; it is 3:1                          |
| 3 : 1   | <b>Non-text:</b> the edge of a control, the focus ring, a meaningful icon | Almost everybody. A page can pass on every word and have fields nobody can find |
| —       | Decorative lines, disabled controls, logos                                | Over-applying it: a 3:1 divider under every card reads as a 1990s table         |

Measure against the colour actually **painted** behind the text, which is often an ancestor's background rather than the element's own.

# THE FOUR STATES, AND HOW EACH ONE FAILS

| STATE   | OWES THE USER                                                        | THE USUAL FAILURE                                                                                             |
|---------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Empty   | What this is, and what to do next                                    | A blank rectangle — or <b>one panel for both empties</b> , so “no results” shows before anything was searched |
| Loading | That something is happening, roughly how much, and a <b>deadline</b> | A spinner with no timeout: it spins until somebody reloads                                                    |
| Error   | What happened · what to do · a way out                               | <b>Error.</b> Or a status code. Or silence                                                                    |
| Success | The data, and <b>where it came from</b>                              | Local fallback data presented as though the network answered                                                  |

In code this is **one field with four values**, not three booleans.

`isLoading + hasError + isEmpty` allows eight combinations, four of which are meaningless — and one of those four will happen.

# FEEDBACK LATENCY: WHAT TO SHOW, AND WHEN

| HOW LONG                 | WHAT THE USER NEEDS                                                                    |
|--------------------------|----------------------------------------------------------------------------------------|
| under 100ms              | Nothing. It feels instant; a spinner here is noise                                     |
| 100ms – 1s               | The result. Do not animate a wait this short                                           |
| 1s – 10s                 | A skeleton or a progress indicator, and the control disabled                           |
| over 10s                 | Progress with an estimate, and a way to cancel                                         |
| <b>8s in this course</b> | <b>Give up and move to the error state.</b> A request with no timeout can hang forever |

Also: a control the user just activated must acknowledge **immediately**, even if the result takes a second. An unacknowledged click gets clicked again.

# VALIDATION TIMING

| EVENT                              | RIGHT FOR                            | WHY                                                                                             |
|------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>input</code>                 | <b>Filtering</b> a local list        | Every keystroke shows more of what you wanted. No wrong intermediate state                      |
| <code>input</code>                 | <b>Never</b> for first validation    | “Not a valid email” after two characters is true, useless, and trains people to ignore messages |
| <code>blur</code>                  | <b>Validation</b> , first time       | They have finished. Now the message is about a state they chose to leave                        |
| <code>input</code> , after invalid | Validation, second time onward       | So the message clears the moment it is fixed, not when they leave again                         |
| <code>submit</code>                | Anything touching the <b>network</b> | Rate limits are real: one endpoint in the pool allows one request per second                    |

# DESTRUCTIVE ACTIONS

## UNDO — THE DEFAULT

Costs nothing up front. Covers the deletion they meant and regretted. Needs a notice that survives long enough to be seen — and focus moved into it.

## CONFIRM — THE EXCEPTION

Only when the action **cannot be undone**. Use a real `<dialog>` with `showModal()`, so the browser traps focus and Escape closes it.

## NEITHER — THE FAILURE

A delete button identical to a cancel button, with nothing between the click and the loss. This scores zero on the rubric row.

A confirmation charges all ninety-nine intentional deletions for the one that was not — and after a week it is clicked through unread.

# THE KEYBOARD CONTRACT

| KEY                    | MUST DO                                                        |
|------------------------|----------------------------------------------------------------|
| <b>Tab / Shift+Tab</b> | Move to the next / previous control, in document order         |
| <b>Enter</b>           | Activate a link or a button. Submit a form from inside a field |
| <b>Space</b>           | Activate a button. Toggle a checkbox. <b>Not</b> a link        |
| <b>Arrows</b>          | Move within a group: radios, a select, a tab list              |
| <b>Escape</b>          | Close whatever opened last, without committing anything        |
| <b>Home / End</b>      | First / last in a group, where a group exists                  |

You get every row of this **for free** from `button`, `a[href]`, `input`, `select` and `fieldset`. Every row you have to write yourself is a row you can get wrong.

# FOCUS MANAGEMENT

- On open** Move focus into the new thing — the dialog, the panel, the first field.
- While open** Keep it in there. `showModal()` does this for you; `show()` does not.
- On close** Put it back where it came from. The half everybody forgets.
- On destroy** Catch it. After a delete the focused element is gone and the browser drops focus to `body`.
- Never** Steal focus on page load, or on a timer. Both throw the user out of whatever they were doing.

## WHERE TO SEND IT AFTER A DELETE

To whatever the user would most plausibly do next — which, after a removal, is **undo**. Not the top of the page.

## POSITIVE TABINDEX

`tabindex="1"` pulls an element to the front of the whole document. Everything stays reachable and the order becomes nonsense. Use `0` or `-1`, never more.

## DOES THIS NEED ARIA?

### 1 Is there a native element for it?

Then use it. `button`, `dialog`, `details`, `fieldset` + radios.

### 2 Is the name already visible text?

Then say nothing. `aria-label` would **replace** it, and the two users are now reading different applications.

### Is it a relationship the DOM cannot express?

### 3 A message belonging to a field: `aria-describedby`. A region that announces changes: `aria-live`.

**This is what ARIA is for.**

### 4 Are you about to add a role to a div?

Stop. It changes the announcement and not the behaviour — no Enter, no Space, no tab stop.

The first rule of ARIA is not to use ARIA. The second is that a wrong role is worse than none: it makes a promise the element does not keep.

# ALT TEXT, AS A DECISION

## 1 Does the image carry information the text does not?

Describe that information. Not the picture — the information.

## 2 Decorative, or a duplicate of adjacent text?

`alt=""` — empty, present, deliberate. A cover beside its own title contributes nothing, and a non-empty alt makes it read twice.

## 3 Is it a link or a button's only content?

The alt **is** the control's name. Describe the destination, not the glyph.

## 4 Is it a chart?

The numbers go in text near it. An alt cannot carry a data set.

### A MISSING ALT IS NOT THE SAME AS AN EMPTY ONE

No `alt` attribute at all makes a screen reader read the **file name**. `alt=""` removes the image from the accessibility tree entirely, which is what you wanted.

# THE THREE SCALES

## TYPE

One ratio, six steps. **1.2 to 1.333** for a page with a dozen elements; wider only for a poster. Under 1.12 is one size with rounding errors.

## SPACE

One step, multiplied.

`--spacing: 0.25rem` gives **4 · 8 · 12 · 16 · 24 · 32 · 48**. Nothing between the steps, ever.

## COLOUR

One hue. Saturation and **lightness** carry the roles — and lightness alone carries the contrast.

A scale is not a restriction; it is a decision taken once so it is not taken forty times. And rhythm is a property of the **set**: nobody can see that 17px is wrong, and everybody can see that 13, 18, 17, 22 has no rhythm.

# MEASURE, AND THE TRAP IN `ch`

| CHARACTERS PER LINE | READS AS                                                           |
|---------------------|--------------------------------------------------------------------|
| under 45            | Work. The eye returns so often that long words break the rag apart |
| <b>45 – 80</b>      | <b>Comfortable.</b> Aim here for anything longer than two lines    |
| over 80             | Tiring. On the way back the eye starts landing on the wrong line   |

`1ch` is the advance width of the digit **zero**, which is wider than the average lowercase letter — so a column set to `62ch` holds noticeably more than 62 characters of prose. `max-width: 62ch` is still the right instrument; just do not read the number as a count.

## TESTING IT ON A PERSON

- 1 One person is enough to find the big thing.**  
Five find almost everything. You do not need a lab.
- 2 Give a task, not a tour.**  
“Register for the Tuesday guitar class.” Then stop talking.
- 3 Do not help.**  
The silence where they are stuck is the finding. Helping destroys the only data you came for.
- 4 Watch the hands, not the face.**  
Where the cursor hovers before it clicks is where they expected the control to be.
- 5 Ask afterwards, never during.**  
“What did you expect that button to do?” — and their answer outranks your intention.

Not examined, and worth an hour anyway: **test your project on one human before week 12.** Whoever is next to you in the lab counts.

# THE FIVE MISCONCEPTIONS

## IT PASSES THE CONTRAST CHECKER

So one row of fifty-five is green. The checker looked at `text` — it has no opinion about the edge of a field, the focus ring, or whether colour is the only cue.

## WHITE SPACE IS WASTED SPACE

It is the cheapest instrument you own. Grouping is a **ratio** between inside and between — and 8 and 16 read exactly like 16 and 32.

## `aria-label` MAKES IT ACCESSIBLE

It **replaces** the visible name. Put one on a button that already says “Send” and the two users are reading different applications. ARIA changes the announcement, never the behaviour.

## VALIDATE AS THEY TYPE

Fourteen errors while somebody types one valid address. Every one true, every one about a state they were always going to pass through. Filter on `input`; validate on `blur`.

## A SPINNER IS THE LOADING STATE

It is half of it. The other half is **what happens when it does not end**. No timeout means the spinner is the final state.

## BONUS: `outline: none` LOOKS CLEANER

It removes the only thing telling a keyboard user where they are. If the default is ugly, **replace** it — `:focus-visible` fires for the keyboard and not for a mouse click, which is what you actually wanted.

# WHERE TO LOOK, IN ORDER

1

## The checklist

`audit-checklist-he.pdf` — fifty-five rows, printable, with how to check each one.

2

## The demos

`demos/01` ... `08` . Every one measures what it claims; open them again while you build.

3

## The brief

`project-brief-he.md` — forty-one requirements, each a criterion you can test.

4

## MDN

for the accessible-name computation and for `dialog` . Both are better documented than they are known.

5

## A person

ten minutes with somebody who has not seen your project beats an hour with any of the above.

# WHAT THE NEXT FIFTY MINUTES GIVE YOU

- 1 A vocabulary**  
hierarchy, proximity, measure, rhythm — words specific enough to act on.
- 2 Three systems**  
a type scale, a palette from one number, a spacing set.
- 3 The four states**  
empty, loading, error, success. The half of every app nobody builds.
- 4 An accessibility method**  
the keyboard, the focus ring, the accessible name.
- 5 A printable checklist**  
fifty-five rows, each with how to check it and what failure looks like.

# THE SAME WORDS, FIVE WAYS

identical markup · only the four levers move

↻ reload

open ↗

THE SAME WORDS, FIVE TIMES

A · nothing

B · size only

C · size, colour inverted

D · size + weight

E · all four

☐ size☐ weight☐ colour☐ position

5-second test

Autumn term

Registration for community classes is now open

Twenty-two classes across five evenings, from beginner yoga to classical guitar. Places are limited and are allocated in the order applications are received, so early registration is recommended for the more popular classes.

Register now

| block          | size | weight | contrast | prominence |
|----------------|------|--------|----------|------------|
| kicker         | 15px | 400    | 17.8:1   | ██████████ |
| title          | 15px | 400    | 17.8:1   | ██████████ |
| body           | 15px | 400    | 17.8:1   | ██████████ |
| call to action | 15px | 400    | 17.8:1   | ██████████ |

No ordering: the heading and the paragraph score the same, so the page has no first thing to read. The outline is still correct — this is invisible to every validator.

► **RUN THE 5-SECOND TEST ON TREATMENT A** and let the silence happen. Then ask which single lever would fix it most.

# HIERARCHY IN THE WILD

## ASOS

“ASOS uses gray for most text but bright green for **Add to Bag** to draw your eyes.”

## COINBASE

“Coinbase highlights **the future of money is here** and their call-to-action **Get Started** to prioritise what they want you to do.”

Both are one decision repeated: **everything else is quiet so that one thing is loud.** Quiet is not a lack of design; it is the design.

# LAYOUT, AND THE LENGTH OF A LINE

## WIKIPEDIA

“Wikipedia pages often have poor layout — very long lines of text, hard to follow.”

## GRAMMARLY

“Grammarly's website has a strong layout — clean spacing, mixed visual elements, clear sections.”

## THE NUMBER

**45–80 characters per line.**

Past 80 the eye starts landing on the wrong line on the way back.

# WHITE SPACE IS NOT EMPTY SPACE

“A luxury handbag in a nearly empty shop window looks more expensive than the same handbag packed in a discount shop.”

Same bag. Same window. The space around an object is part of what the object says.



# ALIGNMENT, AND HOW PEOPLE ACTUALLY READ

## Alignment

Fewer anchor points is cleaner. If titles and paragraphs start in different places, the eye works for nothing.

## F-pattern

Across, then down the leading edge. Text-heavy pages.

## Z-pattern

Diagonal zig-zag. Sparse landing pages.

## Therefore

Headlines and calls to action go where the eye already is.

### RTL CHANGES THE LETTERS, NOT THE RULE

Our pages are right-to-left, so the F starts on the right. Which is exactly why you write `margin-inline-start`, not `margin-left`.

# STOP CHOOSING. START DERIVING.

**1****Type**

one ratio generates every size.

**2****Colour**

one hue generates every colour.

**3****Space**

one step generates every gap.

You already have all three. `tokens.css` in week 3, `@theme` in week 6 — this is the third time the same layer shows up, and the first time you are asked to choose the numbers in it.



# A TYPE SCALE, AND ONE SIZE THAT DOES NOT BELONG

drag the ratio · the measure is COUNTED, not asserted ↶ reload open ↷

THE SCALE — ONE RATIO, SIX SIZES

ratio  1.250 base  16px ☒ include the off-scale heading

|           |        |                          |
|-----------|--------|--------------------------|
| BODY      | 16.0px | Community classes        |
| LEAD      | 20.0px | Community classes        |
| H3        | 25.0px | <b>Community classes</b> |
| OFF-SCALE | 28.1px | <b>Community classes</b> |
| H2        | 31.3px | <b>Community classes</b> |
| H1        | 39.1px | <b>Community classes</b> |
| DISPLAY   | 48.8px | <b>Community classes</b> |

A usable scale — with one size in it that belongs to no step. Look at the column of numbers, not at the specimens: the intruder is obvious in the set and invisible on its own. That is why a rhythm is checked on the page and not on the element.

► **SET THE RATIO TO 1.067** and ask whether it is a scale. Then 1.5. Then find the intruder — in the numbers, not the specimens.

# Every region that shows data has **four** states.

1

## Empty

there is nothing to show.

2

## Loading

we asked and are waiting.

3

## Error

we asked and it failed.

4

## Success

here it is.

You will build the fourth one. **The other three are the half of every student project that does not exist** — and they are half the rubric.

# THE SAME REGION, BADLY AND WELL

one switch drives both columns ↶ reload open ↷

empty (not yet) empty (no matches) loading error success

WHAT USUALLY SHIPS

WHAT THE BRIEF REQUIRES

Search for a class to begin

Type the name of a class or an instructor in the field above. Up to 24 results are shown.

|                                    |      |       |
|------------------------------------|------|-------|
| the state owes the user            | left | right |
| distinguishes the two empty states | no   | yes   |
| tells the user what to do next     | no   | yes   |

► **SWITCH TO LOADING, THEN STOP TALKING.** Take a question about something else and let the spinner run for twenty seconds.

# LOADING OWES TWO THINGS

## A SPINNER

Says “something is happening”. Says nothing about how much, and — with no timeout — spins until somebody reloads.

## A SKELETON

Same shape and size as the real result, so the layout does not jump. Says “three cards are coming”.

## A DEADLINE

**8 seconds, then the error state.** A request with no timeout can hang forever — a captive-portal wifi accepts the connection and never answers.

# AN ERROR STATE OWES THREE

- 1 What happened**  
in a sentence, in their language. Not “Error”, not a status code.
- 2 What to do**  
and “try again” is a real answer when it is a real button.
- 3 A way out**  
a retry, or another route to the same task.

## SCORES 1 OF 3

Error.

## SCORES 3 OF 3

“The catalogue could not be reached. The request timed out after 8 seconds — this is not something you typed.”

**[Try again]** **[Search offline]**

# WHEN TO TELL SOMEONE THEY ARE WRONG

one rule, three timings, a counter on each reload open

**ON EVERY KEYSTROKE — THE "RESPONSIVE" CHOICE**  
Email address  
  
Register  
0 times it said you were wrong

**WHEN YOU LEAVE THE FIELD — WHAT THE BRIEF ASKS FOR**  
Email address  
  
Register  
0 times it said you were wrong

**ONLY WHEN YOU SUBMIT — THE "QUIET" CHOICE**  
Email address  
  
Register  
0 times it said you were wrong

reset the counters

**THE RULE**  
**Filtering** as you type is help: every keystroke shows you more of what you wanted, and there is no wrong intermediate state. **Validating** as you type is nagging: an error about a state the user was always going to pass through, and after a week people stop reading the messages.  
So: local filtering on `input`. Validation on `blur` — then live, but only once the field is already invalid, so the message clears the moment it is fixed rather than making the user leave the field again. Network requests on `submit`, always.

► **ASK A STUDENT TO TYPE THEIR REAL EMAIL** into the first field while the room watches the counter. Explain nothing first.

# DESTRUCTIVE ACTIONS: CONFIRM, OR UNDO

## "ARE YOU SURE?"

Charges all ninety-nine intentional deletions for the one that was not. After a week people click through it without reading.

## UNDO

Costs nothing up front, and also covers the deletion they meant at the time and regretted ten seconds later.

## THE RULE

**Confirm only what cannot be undone.** Almost nothing in your project qualifies.

# THE FOCUS RING IS NOT DECORATION

THE LINE THAT BREAKS A PAGE

```
*:focus {  
  outline: none;  
}
```

IF YOU DISLIKE THE DEFAULT, REPLACE IT

```
*:focus-visible {  
  outline: 3px solid var(--color-brand);  
  outline-offset: 2px;  
}
```

`:focus-visible` is the one you want — it fires for the keyboard and not for a mouse click, which is the actual reason people reach for `outline: none` in the first place.

## A GATE IN THIS REPOSITORY WAS WRONG ABOUT THIS FOR WEEKS

It tested `outline-width`. Chromium **keeps** the width when the style is `none` and simply does not paint it — so a page with every ring stripped reported `3px` and passed. Test the **style**.

# WHERE DOES FOCUS ACTUALLY GO?

the tab path, drawn from document.activeElement ↶ reload open ↷

☐ positive tabindex on three fields ☐ remove the focus ring ☒ manage focus in the dialog clear the path

HOLD TAB. THE PATH IS DRAWN FROM WHERE FOCUS ACTUALLY WENT

Full name

Phone

Email

Class

Beginner yoga ▼

Register Cancel registration...

the stops, in order

Press Tab to begin.

► **SELF-STUDY, DEMOS/08.** Open the dialog with focus management OFF and press Tab six times, narrating what you can see.

# FOCUS MANAGEMENT: THREE MOMENTS

## 1 Something opens

move focus into it. A dialog nobody's focus reached is a dialog nobody can use.

## 2 Something closes

put focus back where it came from. This is the half everybody forgets.

## 3 Something is destroyed

catch it. After a delete, the element that had focus no longer exists.

### WHAT HAPPENS IF YOU SKIP THE THIRD

The browser drops focus to `<body>`. A keyboard user is silently returned to the top of the document and has to tab back through everything; a screen-reader user is told nothing happened at all.

# THE ACCESSIBLE NAME

The browser computes it in this order, and **the first one that produces text wins**.

- 1 · **aria-labelledby** Points at other elements' text.
- 2 · **aria-label** A string you wrote. **It beats the visible text.**
- 3 · **a label** Wrapping the control, or `label[for]`.
- 4 · **alt** · 5 · **own text** For images, then the content — which is what you want, almost always.
- 6 · **title** The last resort, and invisible.

# ARIA ONLY WHERE SEMANTICS FALL SHORT

## FORTY LINES TO ARRIVE BACK WHERE YOU STARTED

```
<div role="radiogroup">
  <div role="radio"
    aria-checked="true"
    tabindex="0">1</div>
  <div role="radio"
    aria-checked="false"
    tabindex="-1">2</div>
</div>
<!-- + keyboard: arrows,
      Home, End, Space -->
```

## THE PLATFORM ALREADY DID IT

```
<fieldset>
  <legend>Rating</legend>
  <label>
    <input type="radio"
      name="r" value="1"> 1
  </label>
  <!-- ...and one per value -->
</fieldset>
```

The right-hand version gets one tab stop for the group, arrow keys inside it, Space to select, the legend as the group's name and “2 of 5” as position — **for free, and correct.**

# ALT TEXT THAT SAYS SOMETHING

## PRESENT, AND WORTHLESS

```
alt="image" ·  
alt="pic2.svg" ·  
alt="תמונה"
```

## DESCRIBES THE CONTRIBUTION

```
alt="The pool, empty,  
with lane ropes in  
place"
```

## CORRECT, AND IT SURPRISES PEOPLE

`alt=""` when the image adds nothing the adjacent text does not already say. **Not a missing alt** — a missing one makes the screen reader read the file name.

# THE SHAPE EVERY PROJECT SHARES

- Three pages** A list · one item · a third page whose every number is **derived**.
- One entity** The collection is made of one kind of thing. “Recipes + ingredients + menus” is three projects.
- Architecture** `state` → `render(state)` → handlers that change state.
- Data** `localStorage` with a defensive load, and one keyless public API with all four states wired.
- Types** One TypeScript module that compiles, in week 12.

## WHY THE SHAPE IS FIXED AND THE TOPIC IS FREE

Week 13 gives everybody the **same three tasks** on forty different applications. That is only possible because they share an architecture.

## WHERE THE POINTS ARE

### 36 — SPEC + ARCHITECTURE

Automatic. Three pages, CRUD, search/filter/sort, the API — and state / render conformance. **The biggest single block, and week 13 rests on it.**

### 24 — STATES + CODE QUALITY

The four states, persistence, defensive load, the TypeScript module, escaping.

### 10 — DESIGN AND UX

Hierarchy, rhythm, consistency, destructive actions. **Ten, not twenty-five.**

Accessibility and responsiveness are 10. The four AI documents and repo hygiene are 12. Beyond-spec is 8. **67 of the 100 points are measured by a machine on the rendered page** — so if you got there another way, you pass.  $36 + 24 + 10 + 10 + 12 + 8 = 100$ .

# A FINISHED ONE, RUNNING

my own build – the same requirements you have

reload open

מדף התקליטים המדף סטטיסטיקה

## המדף שלי

חפש אלבום בקטלוג, הוסף אותו למדף, ותן לו דירוג והערה. הכול נשמר בדפדפן שלך.

### חיפוש בקטלוג

שם אלבום או אמן

חפש

לדוגמה: Radiohead

#### חפש אלבום כדי להתחיל

הקלד שם של אמן או של אלבום בשדה שלמעלה. הקטלוג מחזיר עד 24 תוצאות.

האלבומים שלי

המדף ריק

נקה את המדף

► **SEARCH, ADD, RATE, FILTER, DELETE – THEN PRESS UNDO** and say where the focus went. Then block the network and search again.

← SPINE · 21

38

# THREE LAYERS, AND ONE FILE THAT WIRES THEM

```
1 storage > state > render > screen
2           ^         |
3           +-events--+
4
5 // state   the only place anything is true
6 // render  state in, HTML out. Changes none.
7
8 // events  a click becomes a state change.
9
10 // app     eleven lines, and no more.
```

## THE RULE PEOPLE BREAK FIRST

Anything **derivable is derived**. A `filteredItems` field beside `items` works in every demo and goes stale the first time something forgets to recompute it.

## WHAT IT BUYS YOU

“Make the filter also match the note field” becomes a change to **one pure function** — which is what week 13 asks for.

# SIX WEEKS, AND HOW TO KNOW YOU ARE BEHIND

the end of each week, in your repository

| WEEK | COMMITTED BY THE END OF IT                                      | YOU ARE BEHIND IF...                                   |
|------|-----------------------------------------------------------------|--------------------------------------------------------|
| 7    | topic locked · <a href="#">PROJECT_PLAN.md</a> with section 11  | no topic, or a plan with no field names                |
| 8    | three pages in HTML + Tailwind                                  | one page, or no nav                                    |
| 9    | the UI responds · delegation                                    | buttons do nothing                                     |
| 10   | <a href="#">state</a> + <a href="#">render</a>                  | the list is still written in HTML                      |
| 11   | CRUD + <a href="#">localStorage</a> · the API + all four states | data vanishes on reload, or a spinner and nothing else |
| 12   | TypeScript · keyboard · AA · the four documents                 | <b>submission</b>                                      |

# LIVE, NOW, ON A REAL TOPIC

- 1 Paste the whole template**  
including the guidance comments. They are instructions to it as much as to you.
- 2 Describe the topic in four lines**  
and ask it to mark everywhere it guessed on your behalf.
- 3 Read section 3 first**  
the entity fields. This is where a wrong answer costs two weeks.
- 4 Then section 6**  
the four states, in the words that will appear on screen.
- 5 Fix, write section 11, commit, push**  
and that is part (c) of today's assignment.

► **TAKE A TOPIC FROM THE ROOM.** Do it live, with your mistakes in it. Do not rehearse this one.

# COLOUR ALONE: WCAG 1.4.1, AND WHY AXE CANNOT SEE IT

## THE CRITERION

**1.4.1 Use of Color, level A** — colour is not the only visual means of conveying information, indicating an action, prompting a response, or distinguishing an element. Level A: below AA, below everything.

## WHY THE TOOLS PASS IT

An empty `span` with a background has no text, so there is no contrast to measure and no name to compute. Cycle 1: four such spans, axe at the AA tags reports `color-contrast` on the body text and nothing on them.

## WHAT PASSES IT

A word next to the colour — Places left, Full · waiting list. Test: Rendering, Emulate vision deficiencies, Achromatopsia. If the page still says it, it passes.

# TAILWIND V4: OUTLINE-NONE, OUTLINE-HIDDEN, FOCUS:, FOCUS-VISIBLE:

| UTILITY                     | WHAT IT WRITES (THE VENDORED 4.3.3 BUILD)                                                            | WHEN                                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>outline-none</code>   | <code>outline-style: none</code>                                                                     | Almost never on something focusable. In v3 this name meant what v4 calls <code>outline-hidden</code> . |
| <code>outline-hidden</code> | <code>outline-style: none</code> , and in <code>forced-colors</code> a 2px solid transparent outline | When you replace the ring with something else visible — Windows high contrast still paints one.        |
| <code>focus:</code>         | the <code>:focus</code> pseudo-class                                                                 | Every focus, mouse included.                                                                           |
| <code>focus-visible:</code> | the <code>:focus-visible</code> pseudo-class                                                         | When the browser decides the user needs to see focus — the keyboard. The one for rings.                |

# ROLE="STATUS": A LIVE REGION YOU DECLARE BEFORE IT CHANGES

## What it is

A live region. WAI-ARIA 1.2: `status` has an implicit `aria-live="polite"` and an implicit `aria-atomic="true"`. Chromium's accessibility tree shows it as `status`, live: polite.

## What it does

When its text changes, a screen reader says the new text at the next pause — without moving focus. Polite waits; `assertive` interrupts, and is for errors that cannot wait.

**What it does not do** Announce anything on load, or anything that never changes. Today the line is static; week 9's code will change it.

## The trap

A region created at the moment of the message is often not announced — the reader has to know the region before its content changes. Put it in the markup first.



# WHAT THE AUTOMATIC PLAN CHECKS CAN AND CANNOT SEE

| THE DRAFT                                                 | CORE FEATURES | SERVER OR LOGIN                  | ENTITIES | NULLABLE FIELDS | AUTOMATIC, PART (C) |
|-----------------------------------------------------------|---------------|----------------------------------|----------|-----------------|---------------------|
| P1 — "write me a complete plan"                           | 12            | yes, Node.js + Express + MongoDB | 3        | 0               | 11 / 14             |
| P2 — the template, the brief's constraints, [guess] marks | 5             | only under "will not build"      | 1        | 2               | 14 / 14             |

A machine checks that the plan has a name, ten sections that are not empty, 250 words of yours and a typed entity. It cannot check that the project is this one. **That is 21 manual points and section 11.**